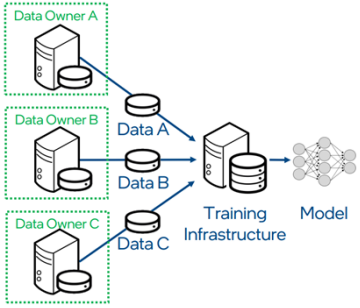
Optimization for Deep Learning (OPT4DL)

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Privacy-Preserving DL

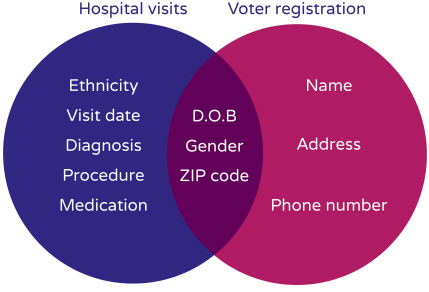
Data Integrity in Distributed DL

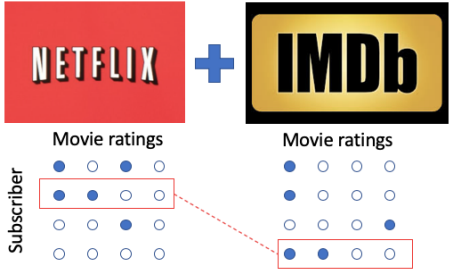
- Distributed
  learning in general leads to performance improvement
- However, information sharing opens the door to
  the possibility of sabotaging the security of personal data
- Ensuring data integrity (a
  context, hence an example of continentally constrained DL)



The need for Randomization

- Anonymization such as removing personally identifiers from data does not work due to linkage attacks (matching anonymized and non-anonymized data)
- Example: Netflix prize
- We need randomized safeguarding mechanisms
- Idea is to learn something useful about a population while not memorizing/learning something specific about an individual agent





Randomized Response



- Assume a survey is being conducted to determine the fraction of professors playing video games
- Assume professors playing video games is considered lame, which of course is not
- So, we want to learn something useful (the fraction) about a population (professors) while not memorizing/learning something specific about an individual agent (whether "professor Hashemi" plays a game)
- Let $x_i \in \{1, 0\}$ denote the truth about whether professor i plays games or not. And our goal is to estimate $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ for a population of n professors (e.g., $n = 120$ in Purdue ECE).
- Assume professor i will reveal an answer $y_i \in \{1, 0\}$ indicating whether they play games or not.
- Case 1: $y_i = x_i$
  - We can accurately estimate $\bar{x}$ by finding the average of y_i's
  - No privacy
- Case 2: $y_i = x_i$ with probability $1/2$ and $y_i = 1 - x_i$ with probability $1/2$
  - y_i's are useless and we cannot estimate $\bar{x}$ at all.
  - perfect privacy

Randomized Response (Cont'd)



- How about interpolating these two cases?
- $y_i = x_i$ with probability $1/2 + \varepsilon$ and $y_i = 1$ with probability $1/2 - \varepsilon$ where $0 \leq \varepsilon \leq 0.5$
- So on average when $\varepsilon > 0$, it is more likely to be truthful than lying. And in some sense provides plausible deniability.
- Let us calculate the mean of y_i

$$\mathbb{E}[y_i] = x_i(0.5 + \varepsilon) + (1 - x_i)(0.5 - \varepsilon) = 2\varepsilon x_i + (0.5 - \varepsilon) \tag{1}$$

meaning y_i is a biased estimator of x_i. Meaning, average of y_i may no be a good estimator of $\bar{x}$.

- After receiving the answers, we can form an unbiased estimator

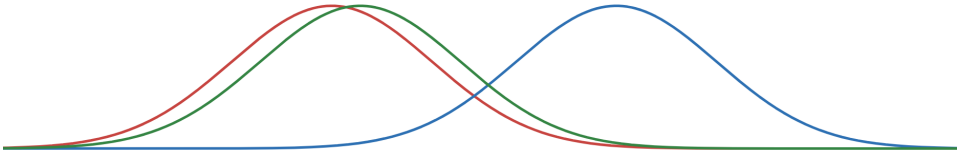
$$\hat{y}_i = \frac{y_i + \varepsilon - 0.5}{2\varepsilon}, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} \hat{y}_i, \quad \mathbb{E}[\bar{y}] = \bar{x}, \quad \mathbb{E}|\bar{y} - \bar{x}| = \mathcal{O}(\frac{0.5 - \varepsilon}{\varepsilon\sqrt{n}}) \tag{2}$$

- Learning something useful ($\bar{x}$) with good accuracy about a population while not memorizing/learning something specific (x_i's) about an individual

Privacy, Sensitivity, and noise

- Assume a professor retires and the department hires a new assistant professor instead.
- Under Case 1 ($y_i = x_i$), our survey is highly sensitive to this change whereas under case 2 ($y_i = x_i$ with probability $1/2$ and $y_i = 1 - x_i$ with probability $1/2$) our survey is completely insensitive.
- Like before noise tends to reduce the sensitivity of our method and in this case improve privacy.
- this idea gives rise to Differential privacy, ensuring

$$\begin{aligned} &\mathbb{P}(\text{ survey (department } \cup \text{ retired)} \in \text{ outcome set)} \\ &\approx \mathbb{P}(\text{ survey (department } \cup \text{ new hire)} \in \text{ outcome set)} \end{aligned} \tag{3}$$



Differential Privacy (DP)

Neighboring datasets

Two datasets $\mathcal{D} \in D_c$ and $\mathcal{D}' \in D_c$ are said to be neighboring if they differ in exactly one sample, and we denote this by $|\mathcal{D} - \mathcal{D}'| = 1$.

(ε, δ)-DP

Given a collection of datasets D_c, a randomized mechanism $\mathcal{M} : D_c \to \mathcal{Y}$ is said to be (ε, δ)-DP, if for any two neighboring datasets and all set of outcomes $\mathcal{R}$

$$\mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{R}) \leq e^{\varepsilon}\mathbb{P}(\mathcal{M}(\mathcal{D}') \in \mathcal{R}) + \delta, \qquad \varepsilon, \delta \geq 0$$

- When $\delta = 0$, it is commonly known as pure DP. Otherwise, it is known as approximate DP.
- e^{ε} is for mathematical convenience. We could have used $1 + \varepsilon$ instead and note $1 + \varepsilon \approx e^{\varepsilon}$
- Symmetric definition despite looking otherwise (note it holds for any two $\mathcal{D}$ and $\mathcal{D}'$)

Differential Privacy (Cont'd)



- Can only be satisfied by randomized mechanism $\mathcal{M}$
- Smaller ε and δ imply stronger privacy guarantees (they control the privacy-accuracy tradeoff)
- The ratio

$$\log \frac{\mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{R})}{\mathbb{P}(\mathcal{M}(\mathcal{D}') \in \mathcal{R})} \tag{4}$$

  is called the privacy loss and DP implies that with probability at least $1 - \delta$, the privacy loss is at most ε
- Typically, $\varepsilon < 0.1$ and $\delta \ll 1/n$ for a problem with n data points to ensure the privacy loss of each data point is small with high probability

Post Processing Theorem

If $\mathcal{M}$ is (ε, δ)-DP, so is its composition with any function h, i.e. $f(\mathcal{M}(\cdot))$.

No matter what others do to the released outcome $\mathcal{M}(\cdot)$, privacy is maintained if $\mathcal{M}(\cdot)$ is a (ε, δ)-DP.

Randomized Response as a DP Mechanism



- Going back to our example on professors and video games, let $\mathcal{D} = \{x_1, \ldots, x_n\}$ and $\mathcal{D}' = \{x_1', \ldots, x_n\}$
- By the Post Processing Theorem, we only need to show our process (i.e. $y_i = x_i$ with probability $1/2 + \varepsilon$ and $y_i = 1$ with probability $1/2 - \varepsilon$ where $0 \leq \varepsilon \leq 0.5$) is DP.
- Our mechanism is then $\mathcal{M}(\mathcal{D}) = [\mathcal{M}_1(x_1), \ldots, \mathcal{M}_n(x_n)]$ where $y_i = \mathcal{M}_i(x_i)$).
- Let $r \in \{0,1\}^n$ be a possible response profile and assume the responses are statistically independent

$$\frac{\mathbb{P}(\mathcal{M}(\mathcal{D}) = r)}{\mathbb{P}(\mathcal{M}(\mathcal{D}') = r)} = \frac{\prod_{i=1}^n \mathbb{P}(\mathcal{M}_i(x_i) = r)}{\mathbb{P}(\mathcal{M}_1(x_1') = r) \prod_{i=2}^n \mathbb{P}(\mathcal{M}_i(x_i) = r)} = \frac{\mathbb{P}(\mathcal{M}_1(x_1) = r)}{\mathbb{P}(\mathcal{M}_1(x_1') = r)} \leq \frac{0.5 + \varepsilon}{0.5 - \varepsilon} \approx e^{5\varepsilon} \tag{5}$$

  for all $\varepsilon \leq 0.3$. Thus, RR is $(5\varepsilon, 0)$-DP.
- Beyond single bit outcomes?

Beyond single bit outcomes



- RR can be thought of as adding the right amount of Bernoulli noise to $x_i \in \{1, 0\}$
- To guarantee DP for broad cases, we can similarly add noise of different kinds to the output.

Laplace Mechanism

Let $h : D_c \to \mathbb{R}^d$ be a function to $\mathbb{R}^d$. The Laplace mechanism is $\mathcal{M}(\mathcal{D}) = h(\mathcal{D}) + [e_1, \ldots, e_d]$ where each $e_i \sim \mathrm{Lap}(0, b)$ is an independent variable distributed according to the Laplace distribution

$$e \sim \mathrm{Lap}(\mu, b) \quad \equiv \quad p(e) \propto \exp(-\frac{|e - \mu|}{b}), \quad \mathbb{E}[e] = \mu, \quad \mathrm{Var}(e) = 2b^2 \tag{6}$$

Gaussian Mechanism

Let $h : D_c \to \mathbb{R}^d$ be a function to $\mathbb{R}^d$. The Gaussian mechanism is $\mathcal{M}(\mathcal{D}) = h(\mathcal{D}) + [e_1, \ldots, e_d]$ where each $e_i \sim \mathcal{N}(0, \sigma^2)$ is an independent variable distributed according to the Gaussian distribution

$$e \sim \mathcal{N}(\mu, \sigma^2) \quad \equiv \quad p(e) \propto \exp(-\frac{|e - \mu|^2}{2\sigma^2}), \quad \mathbb{E}[e] = \mu, \quad \mathrm{Var}(e) = \sigma^2 \tag{7}$$

Sensitivity in Gaussian and Laplace Mechanisms



- Consider Laplace mechanism and let us see how that leads to DP
- Note that in this case $\mathcal{M}(\mathcal{D})$ is a vector where each coordinate is distributed according to $\mathrm{Lap}(h_i(\mathcal{D}), b)$.
- Thus, $\frac{p(\mathcal{M}(\mathcal{D}))}{p(\mathcal{M}(\mathcal{D}'))}$ is simply the ratio of two Laplace distributions with different means
- Simple algebra and the definition of ℓ_1 norm, i.e. $\|x\|_1 = \sum_i x_i$ shows

$$\frac{p(\mathcal{M}(\mathcal{D}))}{p(\mathcal{M}(\mathcal{D}'))} \leq \exp\left(\frac{\|h(\mathcal{D}) - h(\mathcal{D}')\|_1}{b}\right) \tag{8}$$

implying $b \propto \frac{\|h(\mathcal{D}) - h(\mathcal{D}')\|_1}{\varepsilon}$ to ensure $(\varepsilon, 0)$-DP.

Sensitivity in Gaussian and Laplace Mechanisms (Cont'd)



- Thus,
$$b \propto \frac{\|h(\mathcal{D}) - h(\mathcal{D}')\|_1}{\varepsilon} \tag{9}$$
is adequate to ensure $(\varepsilon, 0)$-DP for the Laplace mechanism.
- A similar calculation shows
$$\sigma \propto \frac{\|h(\mathcal{D}) - h(\mathcal{D}')\|_2}{\varepsilon} \log \frac{1}{\delta} \tag{10}$$
to ensure Gaussian mechanism leads to (ε, δ)-DP
- Thus,
$$\max_{\mathcal{D},\mathcal{D}',|\mathcal{D}-\mathcal{D}'|=1} \|h(\mathcal{D}) - h(\mathcal{D}')\|_1, \qquad \max_{\mathcal{D},\mathcal{D}',|\mathcal{D}-\mathcal{D}'|=1} \|h(\mathcal{D}) - h(\mathcal{D}')\|_2 \tag{11}$$
are notions of sensitivity: the more sensitive our learning procedure h, the more noise needed to ensure privacy.
- Implications for DL optimization?

DP in DL Optimization



- Consider our typical learning problem

$$w^* = \arg\min_w f(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i) \tag{12}$$

- Goal is to ensure an approximately optimal model $\hat{w}$ preserves the privacy of data $\mathcal{D} = \{z_1, \ldots, z_n\}$.
- We can leverage DP in three different ways
  - Output perturbation: adding noise to w^*
  - Objective perturbation: adding noise to $f(w)$
  - Gradient Perturbation: leveraging the fact that we use gradient-based methods for DL optimization and adding noise to gradients g_t

Output perturbation



- We hope to simply output $\hat{w} = w^* + e$ where e comes from Laplace or Gaussian dist.
- Unfortunately, The optimal solution w^* can change significantly if a single data point z_i is replaced, particularly when loss has a sharp or degenerate landscape.
- Implication: $\|w^*(\mathcal{D}) - w^*(\mathcal{D}')\|$ could be unbounded and DP cannot be guaranteed
- Under G-Lipschitzness and convexity of $f(w)$, the regularized problem

$$f_\lambda(w) := f(w) + \frac{\lambda \|w\|_2^2}{2} = \frac{1}{n} \sum_{i=1}^{n} \ell(w, z_i) + \frac{\lambda \|w\|_2^2}{2} \tag{13}$$

will be λ-strongly convex and we can show

$$\|w^*(\mathcal{D}) - w^*(\mathcal{D}')\|_2 \leq \frac{2G}{\lambda n} \tag{14}$$

and we can use Gaussian Mechanism

- Two main drawbacks: need convexity and the ability to find the exact solution w^*

Objective Perturbation

- Instead of injecting noise into w^*, inject it into the regularized problem $f_\lambda(w)$ and solve

$$\min_w f_\lambda^e(w) := \frac{1}{n}\sum_{i=1}^n \ell(w, z_i) + \frac{\lambda\|w\|_2^2}{2} + \langle w, e\rangle, \quad e \sim \mathcal{N}(0, \sigma^2 I) \tag{15}$$

- Similar to output perturbation, for guaranteed DP, wee need to compute the optimal w^* exactly
- Let us consider the gradient descent update for such loss function

$$\nabla f_\lambda^e(w) = \frac{1}{n}\sum_{i=1}^n \nabla\ell(w, z_i) + \lambda w + e = \nabla f(w) + \lambda w + e, \qquad w_{t+1} = (1-\eta\lambda)w_t - \eta(\nabla f(w_t) + e) \tag{16}$$

That is, GD with weight decay where all gradients are perturbed with the same Gaussian noise

Gradient Perturbation



- Perturb sample gradients with noise in each iteration (DP-SGD)

$$z_t \sim p_z, \quad , g_t = \nabla \ell(w_t, z_t), \quad e_t \sim \mathcal{N}(0, \sigma^2 I), \quad w_{t+1} = w_t - \eta(g_t + e_t), \tag{17}$$

- We can achieve DP guarantees if $\|\ell(w_t, z_t)\|$ is bounded. True if the sample loss is Lipschitz
- In general, we have to resort to clipping

$$g_t = \nabla \ell(w_t, z_t), \quad \hat{g}_t^C = g_t \min\{1, \frac{C}{\|g_t\|} + \kappa\} \tag{18}$$

or normalization

$$g_t = \nabla \ell(w_t, z_t), \quad \hat{g}_t^N = \frac{C g_t}{\|g_t\| + \kappa} \tag{19}$$

- Both method lead to biased update vectors.
- Can be extended to distributed learning paradigms when each g_t is communicated by a different agent.